RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

CANDIDATE NAME	

CENTRE NUMBER	S					CLASS	21J	INDEX NUMBER					

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions **13 September 2022**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**25**
2	**15**
3	**10**
Section B	**25**
Total	

Section A

Answer **all** the questions in this section.

1 In plants, photosynthesis takes place in chloroplasts, which contain different types of photosynthetic pigments.

(a) Describe the role of different types of photosynthetic pigments in the photoactivation of chlorophyll. [3]

1. Accessory pigments;
2. absorb photons of light;
3. relay energy by resonance to reaction centre;
4. increase the range of wavelength of light absorbed;
5. special pair of chlorophyll a;
6. pass electrons to primary electron acceptor;
7. to initiate electron flow down ETC to drive photophosphorylation;

Scientists suggested that chloroplasts may have originated from prokaryotic cells that continue to function after becoming engulfed by primitive eukaryotic cells.

(b) Other than genomic features, describe **two** evidence that support this hypothesis. [2]

1. Possesses 70S ribosomes;;
2. Chloroplast multiply by binary fission;;
3. Chloroplast surrounded by double membrane;;

Any two

Saline soils, with high concentration of sodium ions (Na^+), are a major problem in many parts of the world. Most plant crop species are unable to tolerate high concentrations of sodium ions.

An experiment was conducted to determine the effect of salt (NaCl) concentration on the overall light-dependent reactions and separately on photophosphorylation activities involving photosystem I and photosystem II.

Table 1.1 shows the results of this experiment on a plant crop species.

Table 1.1

concentration of NaCl / mol dm^{-3}	photosynthetic activity / % activity relative to activity at 0 mol dm^{-3} of NaCl		
	photosystem I	photosystem II	overall light-dependent reactions
0.0	100	100	100
0.1	100	58	80
0.2	98	51	59
0.3	101	35	57
0.4	99	32	39
0.5	100	17	35

Adapted from El-Sheekh MM, Journal of Plant Physiology, 2004

(c) With reference to activities involving the photosystems, predict and explain what will happen to the growth of the plant as salt concentration continues to increase above 0.5 mol dm^{-3}. [3]

1. reduced/no growth;
2. (PSI) cyclic photophosphorylation produces only ATP;
3. reduced non-cyclic photophosphorylation (due to low PSII activity);
4. insufficient NADPH production;
5. for reduction of bisphosphoglylcerate;
6. reduced production of glucose/carbohydrates;
7. no energy for cell division/DNA replication/transcription/translation;

max 3

Recently one group of Na^+ transporters, HKT1, in root cells was found to be important for salt tolerance in plants. HKT1 actively removes Na^+ from the xylem sap. This prevents transport to and accumulation of Na^+ in the shoot.

Scientists have successfully isolated *HKT1* gene from salt-tolerant plants to create a *HKT1* gene construct as shown in Fig. 1.1.

pro	*HKT1*	ter

key
pro – root cell-specific promoter
ter – termination sequence
HKT1 – gene from salt-tolerant plant

Fig. 1.1

HKT1 gene construct was then used to produce salt-tolerant transgenic plant. A transgenic plant refers to a plant whose DNA is modified by inserting a gene from another source.

Fig. 1.2 shows the process of how salt-tolerant transgenic plants is formed.

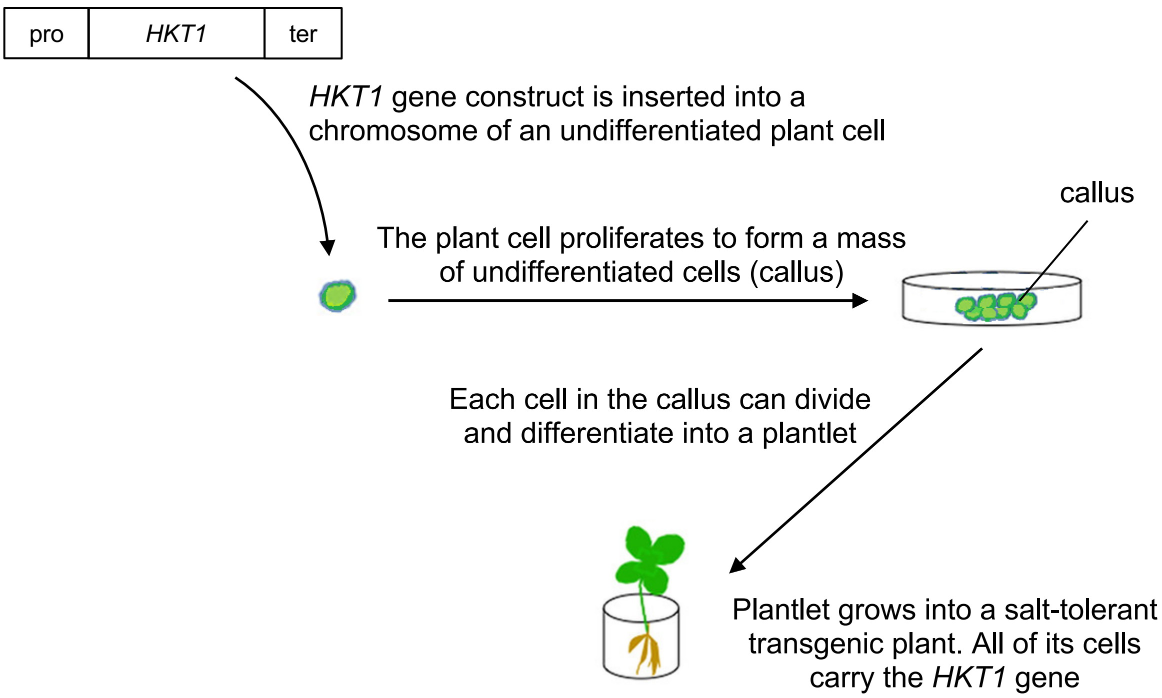

Fig. 1.2

(d) **(i)** Explain why the root cell-specific promoter was included in the *HKT1* gene construct. [2]

1. role of promoter is for binding of RNA polymerase/transcription factors;
2. to initiate transcription <u>*HKT 1* gene</u>;
3. (expression is) only;
4. in the root cells; *Reject: plant cell*

(ii) Cells from the callus in Fig. 1.2 are similar to zygotic stem cells in humans. Explain why these callus cells can be used to produce the entire plant. [2]

1. The cell contains all the DNA/genes/genetic material of the plant;
2. The cell is totipotent;
3. Can divide to form more cells;
4. the cells can differentiate into any cell types;
5. under appropriate condition to form a whole plant;
6. selective gene expression;

Screening was conducted to verify that the 1.7 kb *HKT1* gene has been inserted into the transgenic plant's chromosome. The screening consisted of the following stages:

- DNA was isolated from the transgenic plant.
- The DNA was cut with restriction enzyme and the fragments produced were separated by gel electrophoresis.
- Detection of *HKT1* gene was then carried out.
- Fig. 1.3 shows the region where the radioactive probe binds to on the plant chromosome. The location of restriction site is indicated by *.

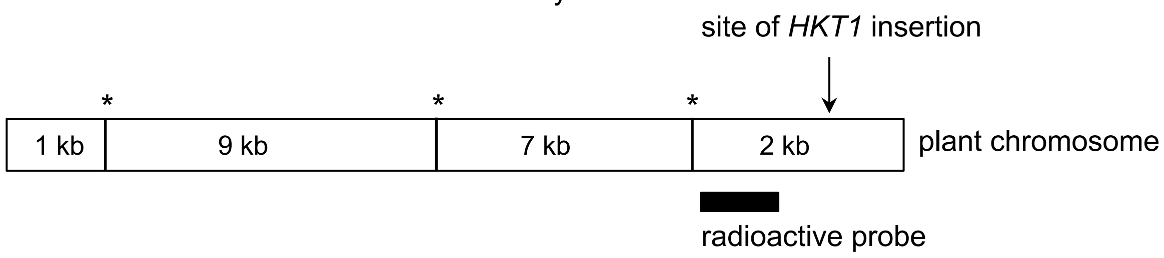

Fig. 1.3

(e) Outline how detection of *HKT1* gene was carried out. [3]

In Southern blotting

1. DNA becomes single stranded;
2. it is denatured by alkaline solution;
3. DNA bands on gel is transferred to nitrocellulose membrane by blotting;

In nucleic acid hybridisation

4. membrane is incubated with radioactive probe;
5. membrane exposed to X-ray film;
6. the size of band correspond to 3.7kb;

An experiment was performed on three groups of transgenic plants (T1, T2 and C) to investigate the effect of *HKT1* gene on plant growth in soils of different salinity. The three groups of plants were as follows:

T1	Transgenic plants containing one copy of the *HKT1* gene
T2	Transgenic plants containing two copies of the *HKT1* gene
C	Control plants without *HKT1* gene

The fresh weight of the plants was determined one week after growing in soils containing different concentrations of NaCl and is shown in Fig.1.4.

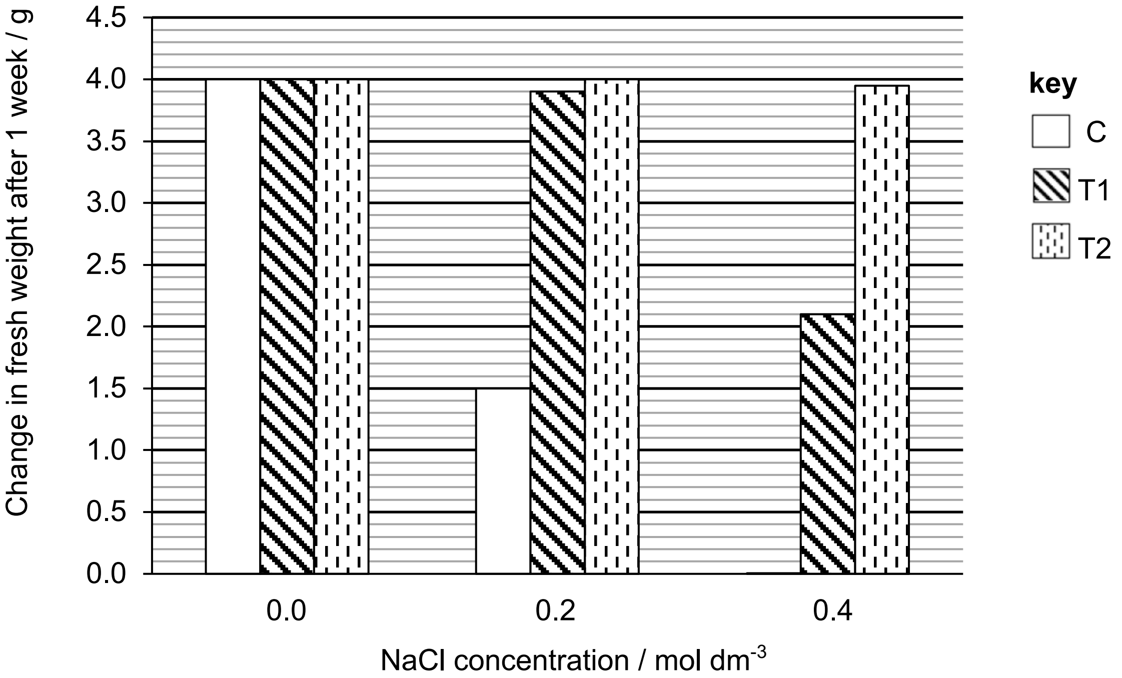

Fig. 1.4

(f) **(i)** With reference to Fig. 1.4, describe the effect of *HKT1* gene on the change in fresh weight of plants at each concentration of NaCl. [3]

At 0 mol dm^{-3} NaCl,

1. Presence of HKT1 gene and increase in number of copies of HKT1 gene has the same change in fresh weight at 4g;;

At 0.2 mol dm^{-3} NaCl,

2. increase in number of copies of HKT1 gene from 1 to 2, similar change in fresh weight at 4g;
3. which is higher than 0 copy of HKT1 at 2g;

At 0.4 moldm^{-3} of NaCl,

4. increase in number of copies of HKT1 gene from 1 to 2, change in fresh weight increases from 2g to 4g;
5. which is higher than 0 copy of HKT1 at 0g;

(ii) explain the difference in the change in fresh weight of T1 and T2 plants. [1]

1. More copies of *HKT1* gene for transcription in T2 than T1;
2. More transporters to remove Na^+;

Fig. 1.5 is a schematic representation of the regions present in the HKT1 transporter, a membrane bound protein.

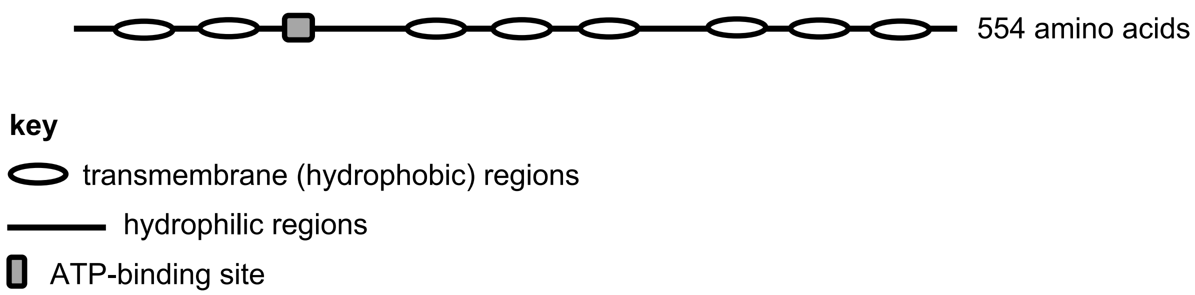

Fig. 1.5

(g) Explain how the different regions shown in Fig. 1.5 allow HKT1 transporter to perform its function. [3]

Structure	Function
Eight transmembrane regions are hydrophobic	form hydrophobic interactions to hold the receptor in the membrane;;
Hydrophilic regions	form the hydrophilic pathway for charged Na^+;;
ATP-binding site	for hydrolysis of ATP to provide energy for active transport/ change shape of transporter;;

Certain plants can adapt to changes in environment by changing the amount of plant hormones, auxin and cytokinin, produced.

Fig. 1.6 shows the effect of drought and mineral nutrient deficiency on auxin and cytokinin biosynthesis in these plants.

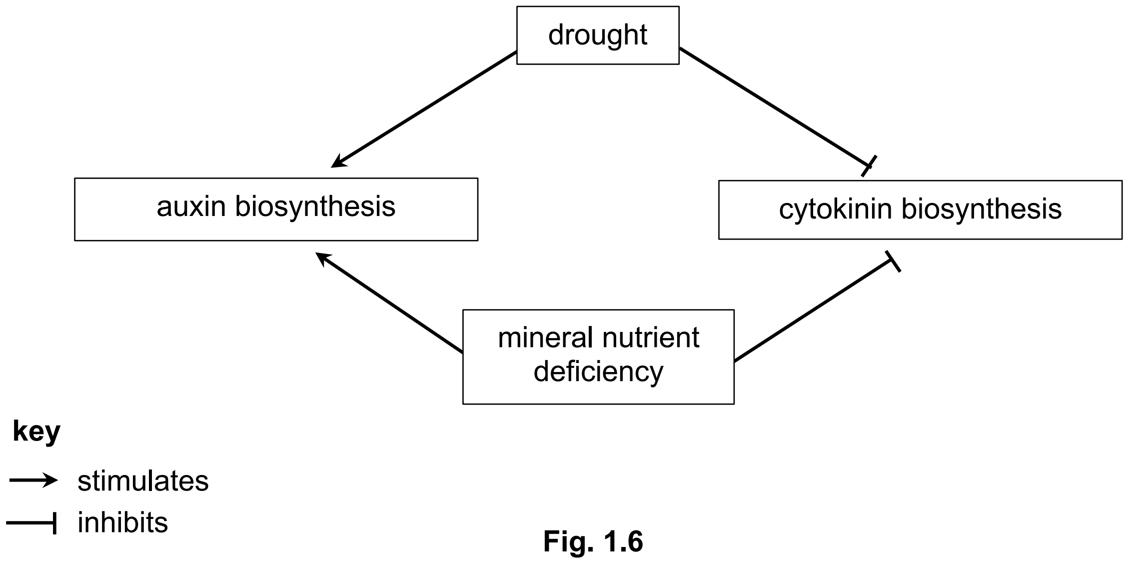

Fig. 1.6

The effects of various ratios of auxin to cytokinin on plant growth are summarised in Table 1.2. The extent of growth is shown on a scale increasing from + to +++.

Table 1.2

ratio of auxin to cytokinin	growth of shoot	growth of root
10:0	no growth	+++
8:2	no growth	++
6:4	no growth	+
5:5	no growth	no growth
4:6	+	no growth
2:8	++	no growth
0:10	+++	no growth

(h) Using Table 1.2 and Fig. 1.6, explain how these plants can survive under conditions of drought and mineral nutrient deficiency. [3]

1. Both inhibit cytokinin biosynthesis;
2. less resources used;
3. no shoot growth;
4. activate auxin biosynthesis;
5. trigger root growth;
6. longer roots to increase water and mineral nutrients absorption from the soil;

[Total: 25]

2 An open reading frame (ORF) is the DNA sequence between a start and stop codon. ORFs are commonly used to predict the presence of genes.

Fig. 2.1 shows the same sample sequence and three possible ORFs due to different possible reading frames. Start codons are underlined and bolded, and stop codons are underlined and italicised.

1 <u>**ATG**</u> CAA TGG GGA AAT GTT ACC CTT ATT GAG GTA AGA CAG ATT <u>*TAA*</u>

2 A TGC ATT GGG GAA <u>**ATG**</u> TTA CCC TTA TTG AGG <u>*TAA*</u> GAC AGA TTT AA

3 AT GCA <u>**ATG**</u> GGG AAA TGT TAC CCT TAT <u>*TGA*</u> GGT AAG ACA GAT TTA A

Fig. 2.1

(a) Using your knowledge of genomic organisation, explain why the use of ORF is more effective in predicting genes in prokaryotic genome than in eukaryotic genome. [1]

In prokaryotic genome

1. Absence of introns;
2. which may contain the triplets coding for start or stop codon;

Accept reverse argument

(b) Outline how the ORF region of sequence **2** may be amplified from a colony of bacteria cells. [4]

1. Extract DNA from cells via homogeniser / lysis;

Perform Polymerase Chain Reaction:

2. At 94°C; denaturation of dsDNA / separation of dsDNA to single-strands;
3. At 65°C; annealing of primers 3'-AAGGGGTTAGC-5'; and 5'-TTAAATCTGTC-3';

   [A: longer primer but within given seq., and also opp. strands]
4. At 72°C; *Taq* polymerase synthesises complementary DNA strand / extension of primers;

max 4

Researchers investigated the possible relationships of ORFs and non-coding sequences with the prokaryote genome size.

Fig. 2.2 shows the relationship between **number of ORFs** and genome size.

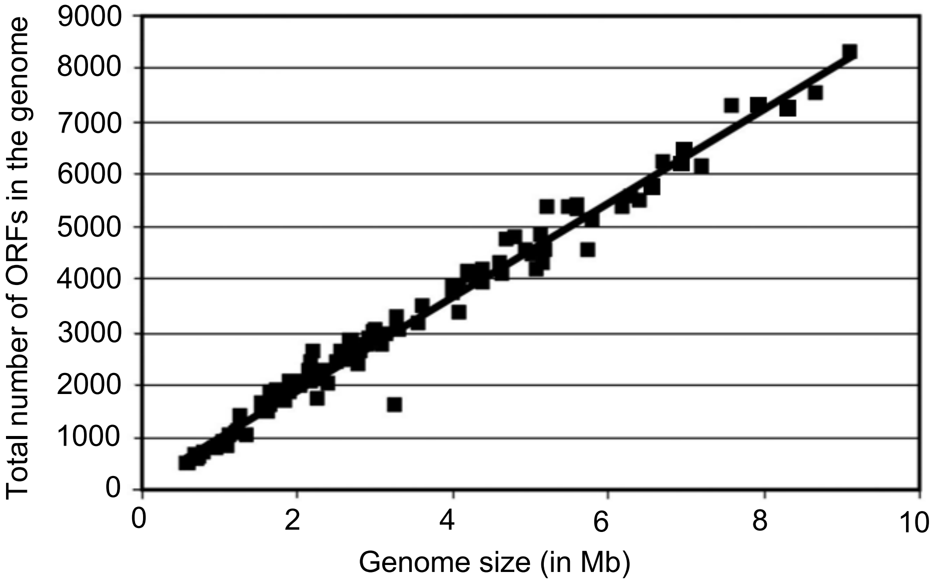

Fig. 2.2

Fig. 2.3 shows the relationship between the **size of non-coding DNA** and genome size.

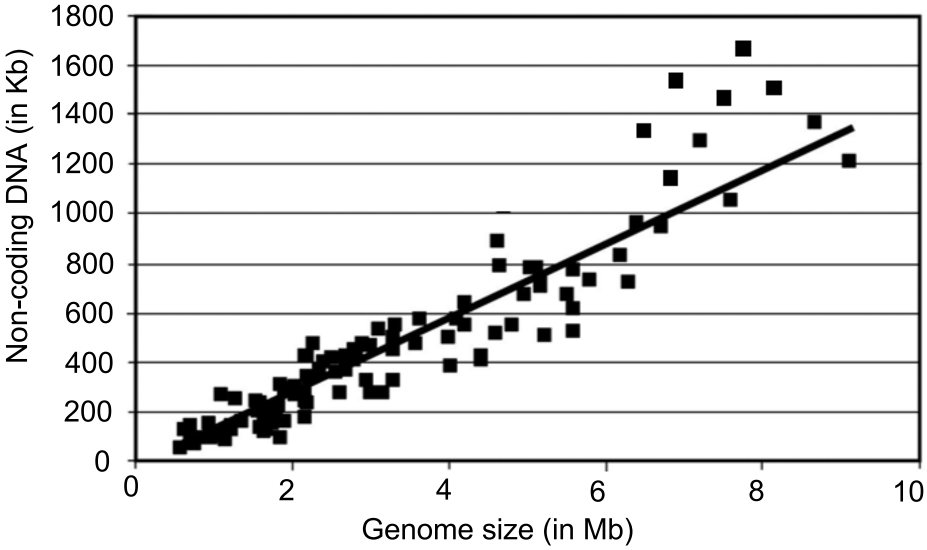

Fig. 2.3

Adapted from Konstantinidis and Tiedje, PNAS, 2004

(c) **(i)** Put a tick (✓) in one box to indicate which relationship shows a weaker correlation between the two variables investigated. [1]

Fig. 2.2		Fig. 2.3	✓

(ii) With reference to your answer in **(c)(i)**, evaluate the extent to which the variable investigated can predict the genome size of the prokaryote. [3]

1. When size of non-coding DNA increases, the genome size increases / reference positive correlation;

2. Prediction at smaller genome size possible due to strong correlation;
3. As data clustered around trendline;

4. But cannot be used to predict genome size for larger genome size / weak correlation for larger genome size;
5. Many data points above trendline;
6. Quote data from Fig. 2.3;;

   e.g. to predict for genome size 8 Mb, should be 1200 Kb non-coding DNA but instead 1500 Kb was present

max 3

Obligate endocellular parasites are one of the smallest genome-sized prokaryotic species.

Fig. 2.4 shows how symbiotic relationship with their host affected genome size in terms of the number of genes and non-coding sequences.

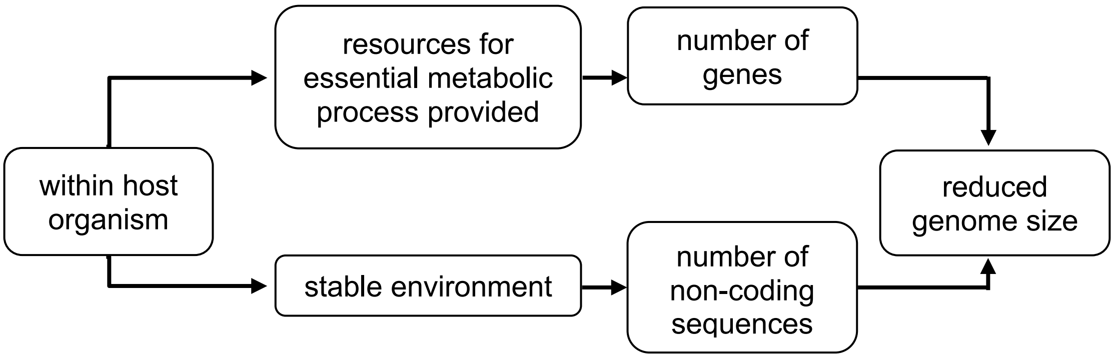

Fig. 2.4

(d) With reference to Fig. 2.4, explain how symbiotic relationship of obligate endocellular prokaryotes with host organisms results in a reduced genome size. [3]

Within the host organism, the prokaryotes

1. Loses genes coding for biosynthesis of resources;
2. Ref resources eg. amino acids, nucleotides etc.;
3. As can be taken up from host;

4. Loses non-coding sequences;
5. Such as (extensive) regulatory sequences;

6. Due to reduced need to respond to changes in environment;

Compared with RNA viruses, prokaryotes have a lower mutation rate.

(e) **(i)** Name an RNA virus. [1]

Influenza virus / Human immunodeficiency virus;;

(ii) Explain why prokaryotes have a lower mutation rate compared to RNA viruses. [2]

1. Prokaryotes have double-stranded DNA;
2. Able to use other strand as template to repair DNA sequence;
3. DNA polymerase has proof-reading ability;
4. Able to excise and replace incorrect nucleotide added at 3' end;

[Total: 15]

3 Rattlesnakes contain a tailshaker muscle which causes a rattling sound upon contraction.

(a) Explain how oxygen uptake by muscle cells changes when the tailshaker muscle contracts. [3]

1. Oxygen uptake increases;
2. More oxygen needed as final electron acceptor of ETC;
3. to maintain electron flow along ETC;
4. more free energy released to pump proton (from matrix to intermembrane space);
5. generate steeper proton gradient across inner mitochondrial membrane;
6. movement of H^+ coupled to increased phosphorylation of ADP to ATP (producing more ATP for muscle contraction);

An investigation was conducted to study the tailshaker muscle and a mammalian muscle.

Fig. 3.1 shows the rate and sources of ATP supply in the tailshaker muscle and the mammalian muscle during contraction.

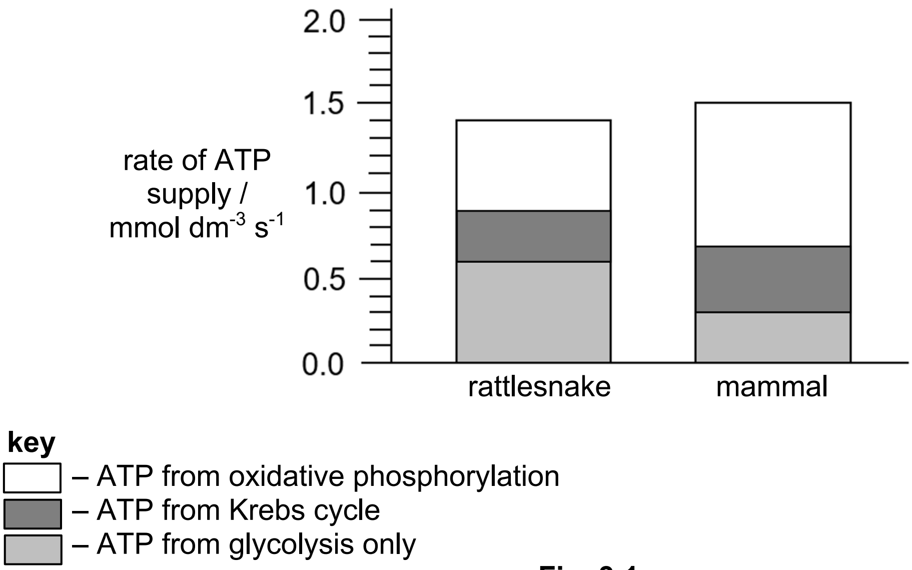

Fig. 3.1

(b) Using Fig. 3.1, calculate the amount of ATP supplied by anaerobic respiration in the tailshaker muscle in 1 minute.

Show your working clearly. [2]

Amount of ATP supplied in tailshaker muscle = 0.6 x 60;;

= 36 mmol dm^{-3};;

Table 3.1 shows the changes in activities in the two muscles during contraction, relative to when the muscles are at rest.

Table 3.1

	change in blood flow / ml min^{-1}	change in O_2 uptake by muscle cells / µmol g^{-1}	change in lactate content in blood / mmol dm^{-3}
tailshaker muscle during rattling	+9.2	+0.148	+2.0
mammalian muscle during contraction	+5.0	+0.180	+5.0

Adapted from Kemper et al., PNAS, 2001

(c) Using the data from Table 3.1, explain why the tailshaker muscle is able to rely on anaerobic respiration for a longer time during rattling as compared to the mammalian muscle. [2]

1. Greater increase in blood flow in tailshaker muscle (+9.2 ml min^{-1}) as compared to that in mammalian muscle (+5.0 ml min^{-1});
2. Lactate produced via lactate fermentation can be transported away quickly;
3. smaller increase in blood lactate content in tailshaker muscle (+2.0 mmol dm^{-3}) as compared to that in mammalian muscle (+5.0 mmol dm^{-3})
4. reduced accumulation of lactate prevents muscle fatigue;

Over the years, climate change has resulted in a rise in mean global temperatures. In Southwestern Europe, scientists collected data on the long-term population trends of a snake species and the rise in temperature in the region from 1980 to 2020, as shown in Fig. 3.2.

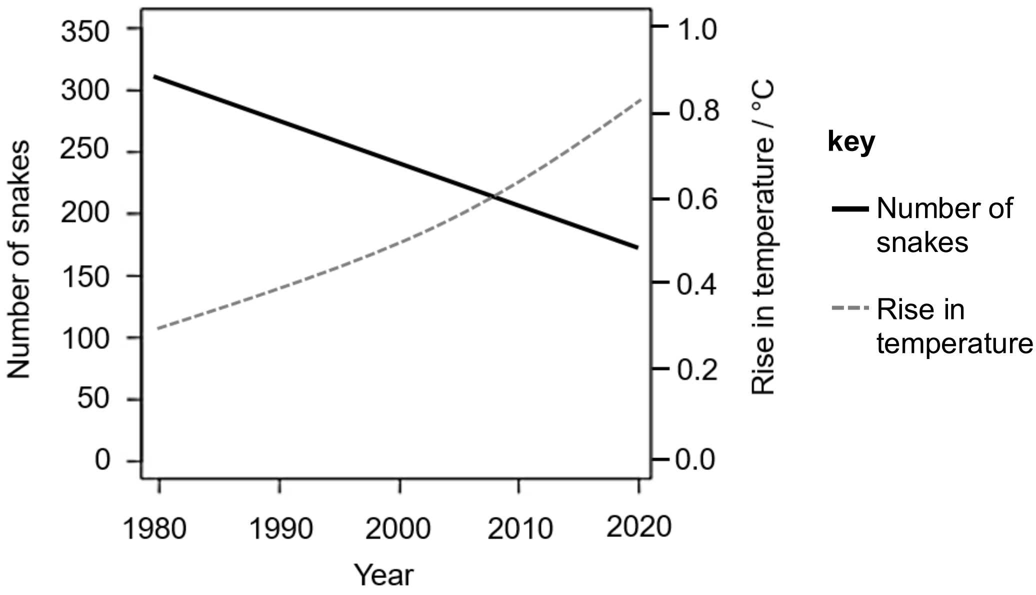

Fig. 3.2

(d) **(i)** With reference to Fig. 3.2, describe the impact of the rise in temperatures in Southwestern Europe on the number of the snake species. [1]

1. As the rise in temperature increases from 0.3°C to 0.82°C (from 1980 to 2020);
2. the relative number of snakes decreased from 310 to 175;

(ii) Suggest reasons for the change in number of snakes from 1980 to 2020. [2]

1. Migration of snakes towards cooler environments where temperatures are optimal for metabolism and survival;;
2. Death of snakes due to high temperatures which denature enzymes essential for survival;;
3. AVP;;

[Total: 10]

Section B

4 **(a)** Outline the functions of proteins involved in DNA replication and DNA condensation to form a chromosome in eukaryotes. [15]

Roles of proteins in DNA replication

1. Helicase;
2. catalyses the breakage of hydrogen bonds between complementary bases;
3. causing the 2 parental DNA strands to separate;
4. single-strand DNA binding proteins;
5. bind to the 2 separated parental DNA strands;
6. to stabilise the single-stranded DNA formed / replication fork;
7. Topoisomerase;
8. creates transient break by nicking a strand of DNA;
9. helps to unwind double helix ahead of replication fork;
10. prevents supercoiling of DNA;
11. Primase;
12. catalyses synthesis of RNA primer in the 5' to 3' direction;
13. DNA polymerase (III);
14. adds nucleotides to the free 3' end of the RNA primer/existing strand;
15. catalyses the formation of phosphodiester bonds between the nucleotides;
16. another DNA polymerase (I);
17. replaces RNA nucleotides of all the RNA primers with DNA nucleotides;
18. DNA ligase;
19. seals the gaps between DNA fragments;

Roles of proteins in packing of DNA to form chromosome

20. Histone proteins;
21. e.g. H2A, H2B, H3 and H4;
22. assemble to form histone octamer / core;
23. upon which DNA is wound around;
24. to form nucleosome;
25. Histone H1;
26. involved in coiling of 10 nm nucleosome fibre;
27. to produce 30 nm chromatin fibre / solenoid;
28. non-histone chromosomal proteins;
29. form scaffold involved in condensing 30 nm chromatin;
30. into looped domains, forming 300 nm fibre;

QWC: Describes roles of at least 2 proteins in DNA replication and 2 proteins in packing of DNA to form chromosomes;;

(b) Describe the structure of viruses and explain the significance of viruses in evolution. [10]

Structure of viruses

1. Contains <u>viral genome</u>;
2. DNA or RNA;
3. linear or circular;
4. single-stranded or double-stranded;
5. can be of positive (+) or negative (-) configuration;
6. Contains <u>capsid</u> (protein coat);
7. comprises large number of protein molecules called capsomeres;
8. arranged in a precise, highly repetitive and highly symmetrical pattern around nucleic acid;
9. helical / icosahedral / more complex shapes;
10. Some viruses contain <u>viral envelope</u>;
11. phospholipid bilayer with embedded glycoproteins;

Significance of viruses in evolution

12. Viruses can causes diseases;
13. e.g. influenza / AIDS / Covid-19 / cancer;
14. **hence act as a selection pressure;**
15. individuals with viral diseases are at a selective disadvantage;
16. will be selected against;
17. healthy individuals are at a selective advantage;
18. will be selected for;
19. can survive and reproduce;
20. and pass on alleles to offspring;
21. increasing allele frequency of normal genes in the population;

22. **Viruses introduce genetic variation;**
23. via bacterial transduction;
24. e.g. generalised / specialised transduction;
25. bacteriophages are transducing phages;
26. carry bacterial genes from one host cell to recipient cell;
27. allow for genetic recombination;

28. **Viruses can cause rapid genetic changes to organisms;**
29. increasing rate of accumulation of mutations;
30. increasing rate at which evolutionary changes occur;

QWC: Describes at least 2 structures of viruses and explains the significance of viruses in evolution;;

[Total: 25]

5 **(a)** Outline the functions of proteins involved in processes which lower high blood glucose concentration. [15]

1. Insulin;
2. as ligand secreted by β-cell of iset of Langerhans;
3. to transmit messages between cells**;**
4. to activate responses in liver cell/muscle cells;

5. Tyrosine kinase receptor (RTK);
6. as receptor on cell surface membrane;
7. for ligand binding to activate signaling pathway;
8. only allow complementary shape ligand to bind to binding site on receptor;
9. determines specificity of signaling pathway;

10. Tyrosine kinase;
11. as enzymes to catalyse phosphorylation / to activate receptor;
12. to phosphorylate tyrosine residues on receptor;

13. kinases;
14. to phosphorylate other kinases in signal transduction;

15. IRS (insulin receptor substrate);
16. as relay protein;
17. to activate other relay proteins;

18. (for kinases and IRS) to trigger phosphorylation cascade;
19. to transduce signal from activated receptors to other proteins in the cell;
20. can result in signal amplification;

21. glucose synthase to convert glucose to glycogen for storage in liver;
22. glucokinase which phosphorylate glucose;
23. glycolytic enzymes to increase rate of glycolysis/ ATP production;
24. enzymes to convert glucose to lipids;
25. enzymes involved in transcription & translation;

26. Glucose transporter;
27. as transport protein;
28. for uptake of glucose;
29. to increase glucose absorption in liver/muscle cells;

30. to lower of blood glucose concentration back to set point;

QWC: Describes at least 3 different functions of proteins;;

(b) Describe how the structure of antibody is related to its function and explain the significance of cell signaling in immune response. [10]

Structure relates to function of antibody

1. (S) Shape of antigen binding sites of antibody is complementary to that of antigen
   (F) bind to and block the interaction of pathogens with the host cell or neutralise their toxins;;
2. (S) Antibody has constant regions on heavy chains

(F) that allows for binding to specific group of effector molecules for antigen elimination;;

3. (S) Antibodies have two antigen binding sites

   (F) for binding to 2 identical structures;;

4. (S) Antibody comprises of antigen binding sites and constant region

   (F) for interactions with antigen and effector cells;;
5. (S) Antibody has variable regions on both heavy and light chains that give rise to a huge variety of antigen binding sites
   (F) for attachment to various antigenic determinant sites;;
6. (S) Antibody has variable heavy chains
7. (F) that gives rise to the 5 classes of antibodies (IgG, IgM, IgA, IgD, IgE) to trigger different effector functions;;

Role of cell signaling in human immune system

1. To activate multiple cells at once;
2. APC activate helper T lymphocyte (via release of interleukin-1);
3. helper T lymphocytes activate B lymphocytes and cytotoxic T lymphocytes (via the release of interleukin-2);

4. To trigger responses of cells;
5. Helper T lymphocytes/B lymphocytes divide by mitosis/clonal expansion;
6. cytotoxic T lymphocytes to kill infected cells;

7. To trigger responses within a cell;
8. helper T lymphocytes to synthesise and secrete cytokinin;
9. plasma cells to synthesise and secrete antibodies (to remove pathogen);

10. Coordination between cells;
11. To integrate and coordinate activities between innate and adaptive immunity;
12. Can attract cells to site of infection through release of cytokinin;

13. To remove pathogen;

QWC: describe at least 2 S-F relationship of antibody and explain the significance of cell signaling in human response;;

[Total: 25]